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Synthetic Memory Note: Inference Keeps Memory Tight

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Synthetic memory demand keeps rising as inference workloads scale across hyperscale clusters, and the operators we spoke with expect the tightness to persist into next year.

Our synthetic channel checks point to HBM supply that stays allocated well ahead of shipments, with contract terms lengthening at every major supplier.

Conventional DRAM contract prices are projected to rise 14% QoQ in 4Q26 in our synthetic model, the third consecutive quarterly increase.

HBM shipments are projected to grow 35% YoY in 2026 as accelerator volumes ramp, while the share of HBM in total DRAM revenue climbs above one third.

Server DRAM inventory at the hyperscalers fell to four weeks in 3Q26, the lowest level in our synthetic series, which leaves little buffer against further demand surprises.

Memory makers' capital expenditure is projected to rise 20% in 2026, with most of the increase directed at advanced DRAM nodes and HBM packaging capacity.

Spot prices for DDR5 modules rose 8% in September in our synthetic tracker, while module makers reported that hyperscale customers were placing orders two quarters ahead of need.

We expect suppliers to keep capacity additions disciplined, because the last downturn left balance sheets stretched and investors rewarded restraint over share gains.

The main risk to this view is a pause in accelerator deployments, which would release HBM capacity back into conventional DRAM and loosen the market within two quarters.